```
#include <stdio.h>
 #include <conio.h>
 #include <stdlib.h>
 int i, j, k, a, b, u, v, n, ne = 1;
 int min, mincost = 0, cost[9][9], parent[9];
 int find(int);
 int uni(int, int);
 void main() {
 printf("\n\tImplementation of Kruskal's Algorithm\n");
 printf("\nEnter the no. of vertices:");
 scanf("%d", & n);
 printf("\nEnter the cost adjacency matrix:\n");
 for (i = 1; i <= n; i++) {
 for (j = 1; j <= n; j++) {
 scanf("%d", & cost[i][j]);
 if (cost[i][j] == 0)
 cost[i][j] = 999;
 }
 }
 printf("The edges of Minimum Cost Spanning Tree are\n");
 while (ne < n) { //edges < vertex - 1
 for (i = 1, min = 999; i <= n; i++) {
 for (j = 1; j <= n; j++) {
 if (cost[i][j] < min) {
 min = cost[i][j];
 a = u = i;
 b = v = j;
 }
 }
 }
 u = find(u);
 v = find(v);
 if (uni(u, v)) {
 printf("%d edge (%d,%d) =%d\n", ne++, a, b, min);
 mincost += min;
 }
 cost[a][b] = cost[b][a] = 999;
 }
 printf("\n\tMinimum cost = %d\n", mincost);
 getch();
 }
 int find(int i) {
 while (parent[i])
 i = parent[i];
```

```
 return i;
 }
 int uni(int i, int j) {
 if (i != j) {
 parent[j] = i;
 return 1;
 }
 return 0;
 }
```

Output:

```
1. Enter the no. of vertices: 4
2.
3. Enter the cost adjacency matrix:
4. 0 10 6 5
5. 10 0 0 15
6. 6 0 0 4
7. 5 15 4 0
8. Enter the no. of vertices: 4
9.
10. Enter the cost adjacency matrix:
11. 0 10 6 5
12. 10 0 0 15
13. 6 0 0 4
14. 5 15 4 0
15.
16. The edges of Minimum Cost Spanning Tree are
17. 1 edge (3,4) =4
18. 2 edge (1,4) =5
19. 3 edge (1,2) =10
20.
21.         Minimum cost = 19
```